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United States Patent	12416138
Kind Code	B2
Date of Patent	September 16, 2025
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Faucet with locking assembly

Abstract

A faucet comprises a faucet body and a locking assembly. The locking assembly is connected to the faucet body, and the locking assembly is configured to pass through a mounting hole of a countertop. At least part of the locking assembly is configured to be controlled to abut a lower side of the countertop, and the faucet body comprises a faucet housing, a support frame, and a valve. The support frame is disposed in the faucet housing, and the support frame is configured to be supported on an upper side of the countertop. The support frame is connected to the locking assembly, and the support frame forms a valve seat configured to be connected to the valve. The locking assembly comprises an operating member rotatably disposed on the support frame, and the operating member is configured to control the locking assembly to abut the lower side of the countertop.

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Appl. No.: 18/533394

Filed: December 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250067031 A1	Feb. 27, 2025

Foreign Application Priority Data

CN	202311072163.6	Aug. 24, 2023
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Publication Classification

Int. Cl.: E03C1/04 (20060101)

U.S. Cl.:

CPC E03C1/0401 (20130101); E03C1/0402 (20130101); E03C1/0403 (20130101); E03C2001/0416 (20130101); Y10T137/6977 (20150401); Y10T137/9464 (20150401)

Field of Classification Search

CPC: E03C (1/0401); E03C (1/0402); E03C (1/0403); E03C (2001/0416); Y10T (137/9464); Y10T (137/6977)

USPC: 137/801; 137/359; 4/695

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to Chinese patent application number 202311072163.6, filed on Aug. 24, 2023. Chinese patent application number 202311072163.6 is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(2) The present disclosure relates to the field of sanitary ware, and in particular to a faucet.

BACKGROUND OF THE DISCLOSURE

(3) Installation of conventional faucets is usually carried out under a countertop, so it requires professionals to bend over for a long time, squat, or lie in a dark and narrow space under a basin and use a variety of special tools to complete the installation. This installation method, which requires going under the countertop for assembly and locking, is time-consuming and laborious for

faucet installers.

(4) A quick installation mechanism for a faucet on a countertop has been disclosed in the prior art, which comprises a mounting base, a screw rod, a nut, a lifting base that rises and falls as the screw rod and nut rotate relative to each other, and two lever rotating arms. The mounting base comprises an installation hole, and the screw rod is disposed on one side of the installation hole. The two lever rotating arms are pivotally connected to the lifting base. This method of installing the faucet from above the countertop makes the faucet assembly simple, and the faucet housing of the faucet can be installed to the mounting base after the mounting base and the lifting base are fixedly connected to the countertop, so it can be applied to faucets of various structures. In addition, this method only uses a single screw, occupying the smallest space in the installation hole, so that a faucet hose can pass through the installation hole smoothly. This mechanism can be used not only in the installation of the faucet on the countertop, but is also suitable for various styles of faucets and can provide a large installation space. It has the advantages of simple structure, reliability, small installation space occupied, and low cost.

(5) However, the quick installation mechanism needs to be tightly connected to the faucet housing. Since the faucet housing needs to play a supporting role, lower strength materials or thin-walled structures cannot be used, and the overall manufacturing cost is high.

BRIEF SUMMARY OF THE DISCLOSURE

(6) The present disclosure provides a faucet to solve the deficiencies in the background.

(7) In order to solve the technical problem, a first technical solution of the present disclosure is as follows.

(8) In order to solve the above technical problems, a faucet is provided. The faucet comprises a faucet body and a locking assembly. The locking assembly is connected to the faucet body, and the locking assembly is configured to pass through a mounting hole of a countertop. At least part of the locking assembly is configured to be controlled to abut a lower side of the countertop, and the faucet body comprises a faucet housing, a support frame, and a valve. The support frame is disposed in the faucet housing, and the support frame is configured to be supported on an upper side of the countertop. The support frame is connected to the locking assembly, and the support frame forms a valve seat configured to be connected to the valve. The locking assembly comprises an operating member rotatably disposed on the support frame, and the operating member is configured to control the locking assembly to abut the lower side of the countertop.

(9) In a preferred embodiment, the locking assembly comprises a guiding base, and the guiding base is connected to a lower end of the support frame. The guiding base comprises a guiding wall extending along a vertical direction, and the guiding wall is configured to divide a lower space located under the support frame into a first space and a second space. The support frame comprises an operation space corresponding to the first space and a pipeline space corresponding to the second space, and the operating member is disposed in the first space and the operation space. A pipeline structure of the faucet is configured to pass through the second space and the pipeline space to be connected to the valve seat.

(10) In a preferred embodiment, the support frame comprises a support base and a support body, and the support base is configured to be supported on the upper side of the countertop and is fixedly connected to the guiding base. The faucet comprises a pipeline pressing plate, and the support body comprises one or more pipeline sockets. The one or more pipeline sockets are in communication with the valve seat, and the pipeline pressing plate is configured to clamp the pipeline structure. The support body is fixedly connected to the support base to fixedly clamp the pipeline pressing plate, and when the support body is connected to the support base, the pipeline structure is correspondingly disposed in the one or more pipeline sockets.

(11) In a preferred embodiment, the valve seat is formed on a side wall of the support frame, and the faucet comprises a valve sleeve. The valve sleeve is connected to the valve seat to form a valve cavity to accommodate the valve, and the faucet comprises a pressing lid. The pressing lid is

connected to a distal end of the valve sleeve to fixedly press the valve.

(12) In a preferred embodiment, the operating member is tubular, and an upper end of the operating member extends out of an upper end of the support frame. The upper end of the operating member is connected to an operating connector, and the operating member is hung on the faucet housing through a neck member to be configured to be rotated on the faucet housing.

(13) In a preferred embodiment, the locking assembly comprises a guiding base, and the guiding base is connected to a lower end of the support frame. The guiding base comprises a guiding wall extending along a vertical direction, and the locking assembly comprises a connecting seat and one or more arms connected to two sides of the connecting seat. One side wall of the connecting seat is in contact with and matched with the guiding wall, and the operating member is threadedly connected to the connecting seat. The operating member is configured to be rotated to drive the connecting seat to move along the guiding wall so that the one or more arms abut the lower side of the countertop.

(14) In a preferred embodiment, the guiding base further comprises a blocking seat connected to the guiding wall, and the one or more arms are configured to be in an unfolded state and a folded state relative to the connecting seat. The one or more arms are controlled by the operating member to approach or move away from the blocking seat, and the one or more arms are in the folded state when the one or more arms are adjacent to the blocking seat. The one or more arms are in the unfolded state when the one or more arms are moved away from the blocking seat, and a radial dimension of the one or more arms in the unfolded state is larger than the mounting hole of the countertop.

(15) In a preferred embodiment, the one or more arms are rotatably connected to the connecting seat, and the one or more arms are divided into a long section and a short section relative to a rotation point of the one or more arms. A length of the long section is longer than a length of the short section, and the connecting seat comprises a protrusion located at a position of the connecting seat adjacent to the long section. The long section is turned over by gravity to drive a lower side of the long section to abut the protrusion so that the one or more arms are in the unfolded state, and the blocking seat comprises a blocking block. When the connecting seat is moved to be adjacent to the blocking seat, the blocking block pushes the long section so that the one or more arms are in the folded state. When the blocking block abuts the long section in the unfolded state, a distance between the blocking block and the rotation point of the one or more arms is greater than a distance between the protrusion and the rotation point of the one or more arms.

(16) In a preferred embodiment, a lower side of the short section comprises a notch, and when the one or more arms are in the folded state, the protrusion corresponds to and is disposed in the notch.

(17) In a preferred embodiment, the one or more arms are rotatably connected to the connecting seat, and the one or more arms are divided into a long section and a short section relative to a rotation point of the one or more arms. A length of the long section is longer than a length of the short section, and the connecting seat further comprises a protrusion located at a position of the connecting seat adjacent to the long section. The long section is turned over by gravity to drive an upper side of the short section to abut the protrusion so that the one or more arms are in the unfolded state, and the blocking seat comprises a blocking block. When the connecting seat is moved to be adjacent to the blocking seat, the blocking block pushes against the long section so that the one or more arms are in the folded state.

(18) In a preferred embodiment, the upper side of the short section comprises a notch, and when the one or more arms are in the unfolded state, the protrusion corresponds to and is disposed in the notch.

(19) In a preferred embodiment, the blocking seat further comprises an inclined surface, and when the one or more arms are in the folded state, the short section abuts the inclined surface.

(20) Compared with the existing techniques, the technical solution has the following advantages.

(21) The support frame is disposed in the faucet housing, and the support frame is configured to be

supported on the upper side of the countertop. The support frame is connected to the locking assembly, and the support frame forms the valve seat configured to be connected to the valve. The locking assembly comprises an operating member rotatably disposed on the support frame, and the operating member is configured to control the locking assembly to abut the lower side of countertop. The support frame is disposed in the faucet housing and is respectively connected to the locking assembly and the valve to provide an overall support effect. A support effect of the faucet housing is weakened or the faucet housing does not play a supporting role. The faucet housing can be made of thin-walled materials or plastic, which can effectively reduce costs.

(22) The first space corresponds to the operation space, and the operating member extends through the first space and the operation space. The pipeline structure extends through the second space and the pipeline space, and two parts of the lower space are separated from each other without interfering or affecting each other. This arrangement can effectively prevent the pipeline structure from being entangled or prevent the pipeline structure from affecting operation of the operating member. In this embodiment, the first space and the operation space are connected to form the axial channel, and the operating member is tubular and rotates around an axis of the operating member on the axial channel.

(23) The pipeline pressing plate is fixedly clamped between the upper end surface of the support base and the lower end of the support body. The pipeline structure penetrates through the support base from the lower end of the support base and partially extends out of the upper end surface of the support base, and the entrance of the one or more pipeline sockets is located on the lower end of the support body. When the support body is correspondingly connected to the support base, the pipeline structure is plugged into the one or more pipeline sockets. During installation, the pipeline structure, guiding base, and support base can be fixedly connected together first, and then the support body and the upper end surface of the support base can be fixedly connected after being installed on the countertop, so as to synchronize achieving a connection between the pipeline structure and the support base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a faucet and a faucet housing in a preferred embodiment of the present disclosure.
- (2) FIG. 2 is a perspective and exploded view of a locking assembly, a support frame, and other components in a preferred embodiment of the present disclosure.
- (3) FIG. 3 is a perspective and exploded view of a support base, a support body, a pipeline structure, the locking assembly, and other components in a preferred embodiment of the present disclosure.
- (4) FIG. 4 is a cross-sectional view of the faucet in a preferred embodiment of the present disclosure.
- (5) FIG. 5 is a perspective view of the faucet in a preferred embodiment of the present disclosure, showing one or more arms in a folded state.
- (6) FIG. 6 is a perspective view of the faucet in a preferred embodiment of the present disclosure, showing the one or more arms in the folded state and about to be installed to a mounting hole.
- (7) FIG. 7 is a perspective view of the faucet in the preferred embodiment of the present disclosure, showing the one or more arms having just been in an unfolded state.
- (8) FIG. 8 is a perspective view of the faucet in the preferred embodiment of the present disclosure, showing the one or more arms are in the unfolded state and abutting a lower side of a countertop.
- (9) FIG. 9 is a perspective view of a connecting seat and the one or more arms in an alternative embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(10) The following will clearly and completely describe the technical solutions in the embodiments of the present disclosure with reference to the accompanying drawings. Obviously, the described embodiments are only a portion of the embodiments of the present disclosure, and not all of the embodiments. Based on the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative work fall within the protection scope of the present disclosure.

(11) In the description of the present disclosure, it should be noted that the terms “upper”, “lower”, “inner”, “outer”, “top/bottom”, etc. indicate the orientation or positional relationship based on the orientation shown in the drawings. The positional relationship is only for the convenience of describing the present disclosure and simplifying the description, rather than indicating or implying that the pointed device or element must have a specific orientation, be constructed, and be operated in a specific orientation. Therefore, the positional relationship should not be understood as a limitation of the present disclosure. In addition, the terms “first” and “second” are only used for descriptive purposes and should not be understood as indicating or implying relative importance.

(12) In the description of the present disclosure, it should be noted that the terms “installed”, “provided with”, “sleeved/connected”, “connected”, etc., should be understood broadly. For example, “connected” can be a fixed connection, a detachable connection, or an integral connection, a mechanical connection, an electrical connection, a direct connection, or an indirect connection through an intermediate medium, and it can be a connection between two members. For those of ordinary skill in the art, the specific meaning of the above terms in the present disclosure can be understood under specific conditions.

(13) Referring to FIGS. **1-8**, a faucet comprises a faucet body **1** and a locking assembly **2**. The locking assembly **2** is connected to the faucet body **1**, and the locking assembly **2** is configured to pass through a mounting hole **201** of a countertop **20**. At least part of the locking assembly **2** is configured to be controlled to abut a lower side of the countertop **20**.

(14) The countertop **20** may be a worktop of a kitchen sink, etc., but the disclosure is not limited thereto. The countertop **20** comprises the mounting hole **201**. The locking assembly **2** can pass through the mounting hole **201** to correspond to the lower side of the countertop **20**. A user can control the locking assembly **2** above the countertop **20** to enable the locking assembly **2** to abut the lower side of the countertop **20** to achieve installation and fixation of the faucet.

(15) The faucet body **1** comprises a faucet housing **11**, a support frame **12**, a valve **13**, a pressing lid **14**, a bent tube **15**, and a handle **16**. The faucet housing **11** is a three-way pipe, and a lower end of the faucet housing **11** is configured to abut an upper side of the countertop **20**. The faucet housing **11** comprises a lateral opening **111**, and the valve **13** is installed into the faucet housing **11** from the lateral opening **111**. A lower end of the bent tube **15** is rotatably connected to an upper end of the faucet housing **11**. A distal end of the bent tube **15** is disposed with a water outlet head **151**. The water outlet head **151** can be a pull-out head or be fixedly connected to the distal end of the bent tube **15**, but the disclosure is not limited to this regard.

(16) The support frame **12** is disposed in the faucet housing **11**. The support frame **12** is configured to be supported above the countertop **20** (i.e., the upper side of the countertop **20**). The support frame **12** is connected to the locking assembly **2**. The support frame **12** forms a valve seat **121** to be connected to the valve **13**. The locking assembly **2** further comprises an operating member **21** rotatably disposed on the support frame **12**. The operating member **21** is configured to control the locking assembly **2** to abut the lower side of the countertop **20**. The support frame **12** is disposed in the faucet housing **11** and is respectively connected to the locking assembly **2** and the valve **13** to provide an overall supporting effect. As a result, a support effect of the faucet housing **11** is weakened or the faucet housing **11** does not provide the support effect. The faucet housing **11** can be made of thin-walled material or plastic, which can effectively reduce costs.

(17) The locking assembly **2** comprises a guiding base **22**, which is connected to a lower end of the

support frame **12**. The guiding base **22** comprises a guiding wall **221** extending along a vertical direction. The guiding wall **221** is configured to divide a lower space located under the support frame **12** into a first space **31** and a second space **32**. The support frame **12** comprises an operation space **122** corresponding to the first space **31** and a pipeline space **123** corresponding to the second space **32**. The operating member **21** is disposed in the first space **31** and the operation space **122**, and a pipeline structure **6** of the faucet passes through the second space **32** and the pipeline space **123** to be connected to the valve seat **121**. The first space **31** corresponds to the operation space **122**, the operating member **21** extends through the first space **31** and the operation space **122**, and the pipeline structure **6** extends through the second space **32** and the pipeline space **123**. The two parts of the lower space are separated and do not interfere with or affect each other, which can effectively prevent the pipeline structure **6** from being entangled or preventing the pipeline structure **6** from affecting operation of the operating member **21**. In this embodiment, the first space **31** and the operation space **122** are connected together to form an axial channel, and the operating member **21** is tubular and rotates around an axis of the operating member **21** on the axial channel.

(18) The support frame **12** comprises a support base **124** and a support body **125**. The support base **124** is configured to be supported above the countertop **20** and is fixedly connected to the guiding base **22**. The faucet further comprises a pipeline pressing plate **4**. The support body **125** comprises one or more pipeline sockets **1251**, which are in communication with the valve seat **121**. The pipeline pressing plate **4** is configured to clamp the pipeline structure **6**. The support body **125** is fixedly connected to the support base **124** to fixedly clamp the pipeline pressing plate **4**. When the support body **125** is connected to the support base **124**, the pipeline structure **6** is correspondingly inserted into the one or more pipeline sockets **1251**. The support base **124** and the support body **125** are arranged separately, and the pipeline pressing plate **4** is clamped between the support base **124** and the support body **125**, which facilitates installation of the pipeline structure **6**. The pipeline structure **6** is inserted into the one or more pipeline sockets **1251**, which is easy and quick to install. Specifically, a lower end surface of the support base **124** and the guiding base **22** are tightly connected through one or more screws **5**, and an upper end surface of the support base **124** and the lower end of the support body **125** are tightly connected through the one or more screws **5**. The pipeline pressing plate **4** is fixedly clamped between the upper end surface of the support base **124** and the lower end of the support body **125**, and the pipeline structure **6** penetrates through the support base **124** from the lower end surface of the support base **124** and partially extends out of the upper end surface of the support base **124**. An entrance of the one or more pipeline sockets **1251** is located on the lower end of the support body **125**, and when the support body **125** is correspondingly connected to the support base **124**, the pipeline structure **6** is inserted into the one or more pipeline sockets **1251**. During installation, the pipeline structure **6**, the guiding base **22**, and the support base **124** can be fixed together first, and then the support body **125** and the upper end surface of the support base **124** can be fixedly connected after being installed on the countertop **20**, thereby synchronously achieving a connection between the pipeline structure **6** and the valve seat **121**.

(19) In this embodiment, the one or more screws **5** are two screws **5**. The two screws **5** connect the guiding base **22**, the support base **124**, and the support body **125** from bottom to top.

(20) The valve seat **121** is formed on a side wall of the support frame **12**. The faucet further comprises a valve sleeve **17**. The valve sleeve **17** is connected to the valve seat **121** to form a valve cavity that accommodates the valve **13**. The pressing lid **14** is connected to a distal end of the valve sleeve **17** to fixedly press the valve **13**. An outer wall of the valve seat **121** comprises external threads, and internal threads at one end of the valve sleeve **17** are connected to the external threads. The pressing lid **14** is connected to internal threads at a second end of the valve sleeve **17**. The pressing lid **14** is not connected to the faucet housing **11** and does not affect the faucet housing **11**.

(21) An upper end of the operating member **21** extends out of an upper end of the support frame **12**.

The upper end of the operating member **21** is connected to an operating connector **23**, and the operating member **21** is hung on the faucet housing **11** through a neck member **24** to be configured to be rotated on the faucet housing **11**. The bent tube **15** can be sleeved outside the upper end of the operating member **21**. The bent tube **15** is connected to the operating member **21** and is not supported on the faucet housing **11**.

(22) The locking assembly **2** comprises a connecting seat **25** and one or more arms **26** connected to two sides of the connecting seat **25**. One side wall of the connecting seat **25** is in contact with and matched with the guiding wall **221**. In this embodiment, the guiding wall **221** is in a concave arc shape, the connecting seat **25** is in a convex arc shape, and the guiding wall **221** and the one side wall of the connecting seat **25** are in surface-to-surface contact. When the operating member **21** is rotated, the connecting seat **25** is limited by the guiding wall **221** to be prevented from rotating to only move up and down along the guiding wall **221**. The operating member **21** is threadably connected to the connecting seat **25**. The operating member **21** can be rotated to drive the connecting seat **25** to move along the guiding wall **221** so that the one or more arms **26** can abut the lower side of the countertop **20**. The guiding base **22** further comprises a blocking seat **222** connected to the guiding wall **221**. The one or more arms **26** are configured to be in an unfolded state and a folded state relative to the connecting seat **25**. The one or more arms **26** are controlled by the operating member **21** to approach or move away from the blocking seat **222**. The one or more arms **26** are in the folded state when the one or more arms **26** are adjacent to the blocking seat **222**, and the one or more arms **26** are in the unfolded state when the one or more arms **26** are moved away from the blocking seat **222**. A radial dimension of the one or more arms **26** in the unfolded state is larger than the mounting hole **201** of the countertop **20**.

(23) The one or more arms **26** are rotatably connected to the connecting seat **25**. The one or more arms **26** are divided into a long section **261** and a short section **262** relative to a rotation point of the one or more arms **26**. A length of the long section **261** is longer than that of the short section **262**. The connecting seat **25** comprises a protrusion **251** located at a position of the connecting seat **25** adjacent to the long section **261**. The long section **261** is turned over by gravity to drive a lower side of the long section **261** to abut the protrusion **251** so that the one or more arms **26** are in the unfolded state. The blocking seat **222** comprises a blocking block **2221**. When the connecting seat **25** is moved to be adjacent to the blocking seat **222**, the blocking block **2221** pushes the long section **261** so that the one or more arms **26** are in the folded state. When the blocking block **2221** abuts the long section **261** in the unfolded state, a distance between the blocking block **2221** and the rotation point of the one or more arms **26** is greater than a distance between the protrusion **251** and the rotation point of the one or more arms **26**. The short section **262** further comprises a notch **2621**. When the one or more arms **26** are in the folded state, the protrusion **251** corresponds to and is disposed in the notch **2621**.

(24) The blocking seat **222** further comprises an inclined surface **2222**. When the one or more arms **26** is in the folded state, the short section **262** contacts the inclined surface **2222**.

(25) Referring to FIG. **9** and FIGS. **1-8**, in some other embodiments, the one or more arms **26** are rotatably connected to the connecting seat **25**, and the one or more arms **26** are divided into the long section **261** and the short section **262** relative to the rotation point of the one or more arms **26**. The length of the long section **261** is longer than that of the short section **262**. The connecting seat **25** further comprises the protrusion **251** located at the position of the connecting seat **25** adjacent to the long section **261**. The long section **261** is turned over by gravity to drive an upper side of the short section **262** to abut the protrusion **251** so that the one or more arms **26** are in the unfolded state. The blocking seat **222** comprises the blocking block **2221**. When the connecting seat **25** is moved to be adjacent to the blocking seat **222**, the blocking block **2221** pushes against the long section **261** so that the one or more arms **26** are in the folded state. The upper side of the short section **262** further comprises the notch **2621**. When the one or more arms **26** are in the unfolded state, the protrusion **251** corresponds to and is disposed in the notch **2621**.

(26) During installation, the guiding base **22**, the support base **124**, the pipeline pressing plate **4**, the pipeline structure **6**, and the support body **125** are connected in the manner described above. The connecting seat **25** is made to abut the blocking seat **222**, so that the one or more arms **26** are in the folded state. The operating member **21** passes through the axial channel and the connecting seat **25** from top to bottom to be connected to a bearing component **2223** on the blocking seat **222**. The operating member **21** can be rotated by manipulation of the operating connector **23** located at the upper end of the operating member **21**. The operating connector **23** is an internal hexagonal socket, and the operation can be achieved by using a tool to be connected to the internal hexagonal socket. The faucet housing **11** is sleeved outside the support frame **12** from top to bottom. The lower end of the faucet housing **11** is positionally sleeved on the support base **124**. The upper end of the faucet housing **11** is connected to an upper end of the support body **125** and the neck member **24** achieves sleeved positioning. The valve sleeve **17** is installed from the lateral opening **111**, and the valve sleeve **17** is threadedly connected to the valve seat **121**. Then the valve **13** is installed on the valve seat **121**, and then fixation of the valve **13** is realized through the pressing lid **14**. The valve **13** is pressed and fixed, and then the handle **16** is connected to the valve **13**. The guiding base **22** passes through the mounting hole **201** from top to bottom, and then the operating member **21** is operated with the tool to rotate until the one or more arms **26** abuts the lower side of the countertop **20**. Finally, the bent tube **15** is connected. In this embodiment, the operating member **21** has a hollow tubular structure. The pipeline structure **6** comprises a pull-out pipe that passes through the operating member **21** and the bent tube **15** to be connected to the water outlet head **151**. The water outlet head **151** is pullably disposed at the distal end of the bent tube **15**.

(27) The aforementioned embodiments are merely some embodiments of the present disclosure, and the scope of the disclosure is not limited thereto. Thus, it is intended that the present disclosure cover any modifications and variations of the presently presented embodiments provided they are made without departing from the appended claims and the specification of the present disclosure.

Claims

1. A faucet, comprising: a faucet body, and a locking assembly, wherein: the locking assembly is connected to the faucet body, the locking assembly is configured to pass through a mounting hole of a countertop, at least part of the locking assembly is configured to be controlled to abut a lower side of the countertop, the faucet body comprises a faucet housing, a support frame, and a valve, the support frame is disposed in the faucet housing, the support frame is configured to be supported on an upper side of the countertop, the support frame is connected to the locking assembly, the support frame forms a valve seat configured to be connected to the valve, the locking assembly comprises an operating member rotatably disposed on the support frame, the operating member is configured to control the locking assembly to abut the lower side of the countertop, the locking assembly comprises a guiding base, the guiding base is connected to a lower end of the support frame, the guiding base comprises a guiding wall extending along a vertical direction, the locking assembly comprises a connecting seat and one or more arms connected to two sides of the connecting seat, one side wall of the connecting seat is in contact with and matched with the guiding wall, the operating member is threadedly connected to the connecting seat, and the operating member is configured to be rotated to drive the connecting seat to move along the guiding wall so that the one or more arms abut the lower side of the countertop.

2. The faucet according to claim 1, wherein: the guiding wall is configured to divide a lower space located under the support frame into a first space and a second space, the support frame comprises an operation space corresponding to the first space and a pipeline space corresponding to the second space, the operating member is disposed in the first space and the operation space, and a pipeline structure of the faucet is configured to pass through the second space and the pipeline space to be connected to the valve seat.

3. The faucet according to claim 2, wherein: the support frame comprises a support base and a support body, the support base is configured to be supported on the upper side of the countertop and is fixedly connected to the guiding base, the faucet comprises a pipeline pressing plate, the support body comprises one or more pipeline sockets, the one or more pipeline sockets are in communication with the valve seat, the pipeline pressing plate is configured to clamp the pipeline structure, the support body is fixedly connected to the support base to fixedly clamp the pipeline pressing plate, and when the support body is connected to the support base, the pipeline structure is correspondingly disposed in the one or more pipeline sockets.
4. The faucet according to claim 1, wherein: the valve seat is formed on a side wall of the support frame, the faucet comprises a valve sleeve, the valve sleeve is connected to the valve seat to form a valve cavity to accommodate the valve, the faucet comprises a pressing lid, and the pressing lid is connected to a distal end of the valve sleeve to fixedly press the valve.
5. The faucet according to claim 1, wherein: the operating member is tubular, an upper end of the operating member extends out of an upper end of the support frame, the upper end of the operating member is connected to an operating connector, and the operating member is hung on the faucet housing through a neck member to be configured to be rotated on the faucet housing.
6. The faucet according to claim 1, wherein: the guiding base further comprises a blocking seat connected to the guiding wall, the one or more arms are configured to be in an unfolded state and a folded state relative to the connecting seat, the one or more arms are controlled by the operating member to approach or move away from the blocking seat, the one or more arms are in the folded state when the one or more arms are adjacent to the blocking seat, the one or more arms are in the unfolded state when the one or more arms are moved away from the blocking seat, and a radial dimension of the one or more arms in the unfolded state is larger than the mounting hole of the countertop.
7. The faucet according to claim 6, wherein: the one or more arms are rotatably connected to the connecting seat, the one or more arms are divided into a long section and a short section relative to a rotation point of the one or more arms, a length of the long section is longer than a length of the short section, the connecting seat comprises a protrusion located at a position of the connecting seat adjacent to the long section, the long section is turned over by gravity to drive a lower side of the long section to abut the protrusion so that the one or more arms are in the unfolded state, the blocking seat comprises a blocking block, when the connecting seat is moved to be adjacent to the blocking seat, the blocking block pushes the long section so that the one or more arms are in the folded state, and when the blocking block abuts the long section in the unfolded state, a distance between the blocking block and the rotation point of the one or more arms is greater than a distance between the protrusion and the rotation point of the one or more arms.
8. The faucet according to claim 7, wherein: a lower side of the short section comprises a notch, and when the one or more arms are in the folded state, the protrusion corresponds to and is disposed in the notch.
9. The faucet according to claim 6, wherein: the one or more arms are rotatably connected to the connecting seat, the one or more arms are divided into a long section and a short section relative to a rotation point of the one or more arms, a length of the long section is longer than a length of the short section, the connecting seat further comprises a protrusion located at a position of the connecting seat adjacent to the long section, the long section is turned over by gravity to drive an upper side of the short section to abut the protrusion so that the one or more arms are in the unfolded state, the blocking seat comprises a blocking block, and when the connecting seat is moved to be adjacent to the blocking seat, the blocking block pushes against the long section so that the one or more arms are in the folded state.
10. The faucet according to claim 9, wherein: the upper side of the short section comprises a notch, and when the one or more arms are in the unfolded state, the protrusion corresponds to and is disposed in the notch.

11. The faucet according to claim 9, wherein: the blocking seat further comprises an inclined surface, and when the one or more arms are in the folded state, the short section abuts the inclined surface.

12. The faucet according to claim 7, wherein: the blocking seat further comprises an inclined surface, and when the one or more arms are in the folded state, the short section abuts the inclined surface.

13. A faucet, comprising: a faucet body, and a locking assembly, wherein: the locking assembly is connected to the faucet body, the locking assembly is configured to pass through a mounting hole of a countertop, at least part of the locking assembly is configured to be controlled to abut a lower side of the countertop, the faucet body comprises a faucet housing, a support frame, and a valve, the support frame is disposed in the faucet housing, the support frame is configured to be supported on an upper side of the countertop, the support frame is connected to the locking assembly, the support frame forms a valve seat configured to be connected to the valve, the locking assembly comprises an operating member rotatably disposed on the support frame, the operating member is configured to control the locking assembly to abut the lower side of the countertop, the locking assembly comprises a guiding base, the guiding base is connected to a lower end of the support frame, the guiding base comprises a guiding wall extending along a vertical direction, the guiding wall is configured to divide a lower space located under the support frame into a first space and a second space, the support frame comprises an operation space corresponding to the first space and a pipeline space corresponding to the second space, the operating member is disposed in the first space and the operation space, and a pipeline structure of the faucet is configured to pass through the second space and the pipeline space to be connected to the valve seat.

14. The faucet according to claim 13, wherein: the support frame comprises a support base and a support body, the support base is configured to be supported on the upper side of the countertop and is fixedly connected to the guiding base, the faucet comprises a pipeline pressing plate, the support body comprises one or more pipeline sockets, the one or more pipeline sockets are in communication with the valve seat, the pipeline pressing plate is configured to clamp the pipeline structure, the support body is fixedly connected to the support base to fixedly clamp the pipeline pressing plate, and when the support body is connected to the support base, the pipeline structure is correspondingly disposed in the one or more pipeline sockets.

15. The faucet according to claim 13, wherein: the valve seat is formed on a side wall of the support frame, the faucet comprises a valve sleeve, the valve sleeve is connected to the valve seat to form a valve cavity to accommodate the valve, the faucet comprises a pressing lid, and the pressing lid is connected to a distal end of the valve sleeve to fixedly press the valve.

16. The faucet according to claim 13, wherein: the operating member is tubular, an upper end of the operating member extends out of an upper end of the support frame, the upper end of the operating member is connected to an operating connector, and the operating member is hung on the faucet housing through a neck member to be configured to be rotated on the faucet housing.

17. A faucet, comprising: a faucet body, a locking assembly, a valve sleeve, and a pressing lid, wherein: the locking assembly is connected to the faucet body, the locking assembly is configured to pass through a mounting hole of a countertop, at least part of the locking assembly is configured to be controlled to abut a lower side of the countertop, the faucet body comprises a faucet housing, a support frame, and a valve, the support frame is disposed in the faucet housing, the support frame is configured to be supported on an upper side of the countertop, the support frame is connected to the locking assembly, the support frame forms a valve seat configured to be connected to the valve, the locking assembly comprises an operating member rotatably disposed on the support frame, the operating member is configured to control the locking assembly to abut the lower side of the countertop, the valve seat is formed on a side wall of the support frame, the valve sleeve is connected to the valve seat to form a valve cavity to accommodate the valve, and the pressing lid is connected to a distal end of the valve sleeve to fixedly press the valve.

18. A faucet, comprising: a faucet body, and a locking assembly, wherein: the locking assembly is connected to the faucet body, the locking assembly is configured to pass through a mounting hole of a countertop, at least part of the locking assembly is configured to be controlled to abut a lower side of the countertop, the faucet body comprises a faucet housing, a support frame, and a valve, the support frame is disposed in the faucet housing, the support frame is configured to be supported on an upper side of the countertop, the support frame is connected to the locking assembly, the support frame forms a valve seat configured to be connected to the valve, the locking assembly comprises an operating member rotatably disposed on the support frame, the operating member is configured to control the locking assembly to abut the lower side of the countertop, the operating member is tubular, an upper end of the operating member extends out of an upper end of the support frame, the upper end of the operating member is connected to an operating connector, and the operating member is hung on the faucet housing through a neck member to be configured to be rotated on the faucet housing.
